Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	Smart Lender – AI Based Loan Eligibility Prediction System
Maximum Marks	5 Marks

Proposed Solution report : The Smart Lender project aims to automate the loan eligibility prediction process using Machine Learning. The system analyzes applicant information such as income, employment status, education, credit history, loan amount, and other financial details to predict whether a loan application is likely to be approved. This solution reduces manual verification, minimizes human errors, improves prediction accuracy, and enables banks to make faster and more consistent lending decisions through a user-friendly web application.

Project Overview	
Objective	To develop a Machine Learning model that accurately predicts loan eligibility based on an applicant's financial and personal information.
Scope	• Collect loan applicant data. • Clean and preprocess the dataset. • Perform exploratory data analysis. • Train and evaluate Machine Learning models. • Predict loan eligibility (Approved/Rejected). • Deploy the model using a Flask web application. • Display prediction results through an interactive user interface.

Problem Statement	
Description	Manual loan approval is time-consuming, error-prone, inconsistent, and unable to efficiently process a large number of loan applications.
Impact	• Delayed loan approvals. • Increased operational costs. • Human bias in decision-making. • Poor customer experience. • Risk of incorrect lending decisions.

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SKILL WALLET A SmartBridge Product

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel i5/Ryzen5
Memory	RAM	8 GB
Storage	Disk space	256 GB SSD
Software		

Frameworks	Backend	Flask
Libraries	Python libraries	scikit-learn, pandas, NumPy, matplotlib,pickle
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Dataset	Source	Kaggle Credit Card Approval Predictions(CSV)